Internship Learning Log

Solar Module Manufacturing — from cell to shipped panel

Prepared by Varish

June 15–19, 2026

Chikli, Vapi

4-Day Summary

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Internship Overview

Dates	June 15–19, 2026
Location	Chikli, Vapi — Gujarat
Mentors	Dr. Prasann Shah & Arjun Patel
Host	Waaree Energies — #1 non-Chinese solar module manufacturer globally
Context	IBDP Year 1 summer internship; bridging classroom physics with industrial-scale engineering reality

MY LEARNING OBJECTIVES

Theory → Practice
Bridge classroom physics to manufacturing-scale reality

Electrical Testing
I-V curves, efficiency measurement & quality gates

Automation & Control
Industrial systems — links to my robotics background

Engineering Culture
How engineers communicate & problem-solve in practice

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About Waaree Energies

22.77 GW
Total module capacity (India + US)

5.4 GW
Solar cell capacity — India's largest

₹26,537 Cr
FY26 revenue (3-yr CAGR 58%)

25+
Countries with active sales

Waaree 1.0
26 GW modules · 5.4 GW cells

→

Waaree 2.0
28 GW modules · 15 GW cells

~$3.5Bn Capex · TAM $1Tn today → $4Tn by 2035

- **#1 globally:** Largest non-Chinese solar module manufacturer
- **1 GW/month:** First Indian co. to hit this milestone (Nov 2025)
- **US presence:** 2.6 GW in Texas & Arizona — benefits from anti-China tariffs
- **Deep capex:** ₹30,000 Cr+ across modules, cells, BESS, hydrogen, glass & T&D
- **Chikli facility:** 5.4 GW TOPCon cells at 25.54% efficiency — ALMM List-II certified

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Day 1 — Orientation & Factory Tour

ENTRY & ORIENTATION

- **Security check-in** — tight clearance before campus entry
- **OHF planning** — 5-day schedule mapped with mentors
- **Safety briefing** — thorough protocol induction
- **Technical induction** — raw materials, process & specs
- **Campus discipline** — strict 20 kmph speed limit observed

FACTORY TOUR

- **PPE mandatory** — safety boots & hairnets for all
- **Forklift awareness** — sudden emergence from box stacks
- **Glass sorting** — incoming sheets inspected before line
- **Cell stringing** — soldering & quality-testing half-cells
- **Back-end** — EVA lamination, trimming, framing, flash test

4-DAY INTERNSHIP PLAN

DAY 1	DAY 2	DAY 3	DAY 4
Module Basics & Back-End Layer structure, cell interconnection, lamination & framing	**Testing & Build Flow** Reliability tests & full cell-to-panel production line	**Incoming Material QC** Sampling & testing every raw material before the line	**Utilities & Raw Materials** Plant power, support equipment & BOM

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Key Observations

1. **Scale that redefines expectations**
   Witnessing 1 GW+ monthly output in practice — not as a statistic — fundamentally changed my sense of what industrial scale means.
2. **Two breeds of engineer**
   Engineers who design machines vs. those who operate them at scale. At Rythmsoft (last year) I saw design; here I saw mass deployment. Both are essential — both fascinated me.
3. **Automation balanced with human judgement**
   Defective cells swapped manually — more efficient than discarding an entire laminated panel. Knowing when to automate and when to intervene is itself a discipline.
4. **Precision at nano scale, deployed at giga scale**
   EL imagers, sun simulators, UTMs — maintaining sub-millimetre tolerances across thousands of panels daily.
5. **The human architecture of 24/7 production**
   Structured shifts, mandatory and voluntary breaks, fatigue management. Operational continuity is as much a people challenge as an engineering one.

What I Learned, Day by Day

Waaree Energies · Chikli, Vapi · June 2026
Prepared by Varish · IBDP Year 1

DAY 1 Module Basics & Back-End Process

How a solar module is layered, interconnected and assembled

The 5-Layer Sandwich

1. **Front Glass — semi-toughened, ARC coated, textured inner face**
2. **Front EVA — bonds to glass, UV-resistant, transparent when melted**
3. **Solar Cell — silicon half-cells; the electricity-generating layer**
4. **Back EVA — shields rear, prevents vibration damage**
5. **Back Glass — bifacial modules capture albedo light from rear**

Tabber & Stringer

Automated machine solders cells into strings before layup. Silver fingers and busbar dots are screen-printed on each cell; copper ribbon solders top-to-bottom alternating to carry current through the string.

16 BB per cell · 12 cells/string · UTM pull test

Lamination → Framing

Laminate: EVA melted at ~150°C to bond all layers. **Vacuum:** air extracted to prevent corrosion & bubbles. **Cool:** panel held near 50°C for even cure. Aluminium frame added with RTV sealant (weighed before & after). Flash/pump test measures actual power output.

DAY 2 Reliability Testing & Manufacturing Flow

Why modules are tested — and how they travel from cell to shipped panel

Electrical & Performance

EL imaging · Sun-simulator · Insulation & wet-leakage · Bypass diode · Impulse voltage · Reverse-current overload

🌡 Environmental & Climate

Damp heat (85°C/85%RH) · Humidity freeze · Thermal cycling +85↔−45°C · UV preconditioning · Salt-mist · LeTID · PID (96h)

Mechanical & Structural

Static load push/pull · Bending · Junction box pull · Module breakage · Peel & lap-shear bonding

🛡 Safety & Handling

Hail impact · Water-leak immersion · Cut susceptibility · Sharp-edge · Blade probe · Visual inspection

Production Line — 8 Steps

① **Cells in** — silicon cells cut into half-cells for lower losses
② **Stringing** — ribbons flux-soldered in alternating top/bottom pattern
③ **EL check** — electroluminescence photo confirms every joint
④ **Layup** — 6 rows per half, 3 alignment holes, tape holds spacing
⑤ **Lamination** — printed glass laminated, vacuum pickers align
⑥ **Inspection** — computer scans; faulty cells swapped in repair area
⑦ **Cure & test** — ~5h cool, sun-simulator check, barcode & logo
⑧ **Pack & ship** — defective pulled; good panels packed and shipped

DAY 3 Incoming Material Quality Checks

Every raw material sampled and tested before it enters the line

Material	What's Checked	Tool / Method
Solar cell	Dimensions, electrical ratings, efficiency, whitespot defects	Vernier callipers; sun simulator
Ribbon & busbar	Diameter, length, tensile strength, elasticity	Vernier; UTM stretch-to-break
Aluminium frame	Dimensions, hardness, anodised coating	Vernier + tape; hardness tester; rust-analysis machine
EVA encapsulant	Thermal stability, tensile strength, lamination shrinkage	Xenon chamber 30h @ ~70°C, then peel test
Glass	Dimensions, transparency, impact strength	Steel-ball drop test
Junction box	Waterproofing of sealed joint	30-min water immersion
Flux	Acidity level	Titration vs hydrochloric acid
Cardboard box	Compression / crush strength	Force-to-collapse — BS testing machine

DAY 4 Plant Utilities & Raw Materials

The power, support equipment and materials behind every panel

Electrical Infrastructure

Three-phase power enters at **11,000 V**, stepped down in the plant control room before reaching the line.

- **MCB** — Miniature CB, low-current branches
- **ACB** — Air CB, larger loads
- **VCB** — Vacuum CB, high-voltage switching

Compressor

rated in CFM

Supplies compressed air to line

Chiller

rated in TR

Process cooling throughout plant

Dryer

Removes humidity from compressed air

AHU

Air handling — filters & conditions

Cooling Tower

Rejects waste heat from loop

Transformer

rated in kVA

Steps voltage down for line use

Bill of Materials

Glass · EVA/EPE/POE · Solar cell · Ribbon · Flux · Busbar · Tape · Aluminium frame · RTV sealant · Junction box · IPA cleaner · Barcode & RFID · Corner protector · Back label